



Lecture 11: Health Care Associated Infections (HAIs)



By

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Intended Learning Outcomes (iLOs)

By the end of this lecture, students will be able to:

1. **Define** the term nosocomial infection.
2. **Identify** the main sources of nosocomial infections.
3. **Explain** factors that increase patient susceptibility to hospital-acquired infections.
4. **Recognize** common bacterial, viral, and fungal pathogens causing nosocomial infections.
5. Outline effective preventive measures for nosocomial infections.

What is a nosocomial infection?

Nosocomial infections also known as hospital-acquired infections (**HAIs**) or healthcare-associated infections (**HCAIs**) are infections that develop **48 hours** or more after a patient is admitted to a hospital or other healthcare facility, and were not present or incubating at the time of admission.

The term nosocomial originates from the Greek word **nosokomeion**, meaning **hospital**.



Pathogens Causing Nosocomial Infection

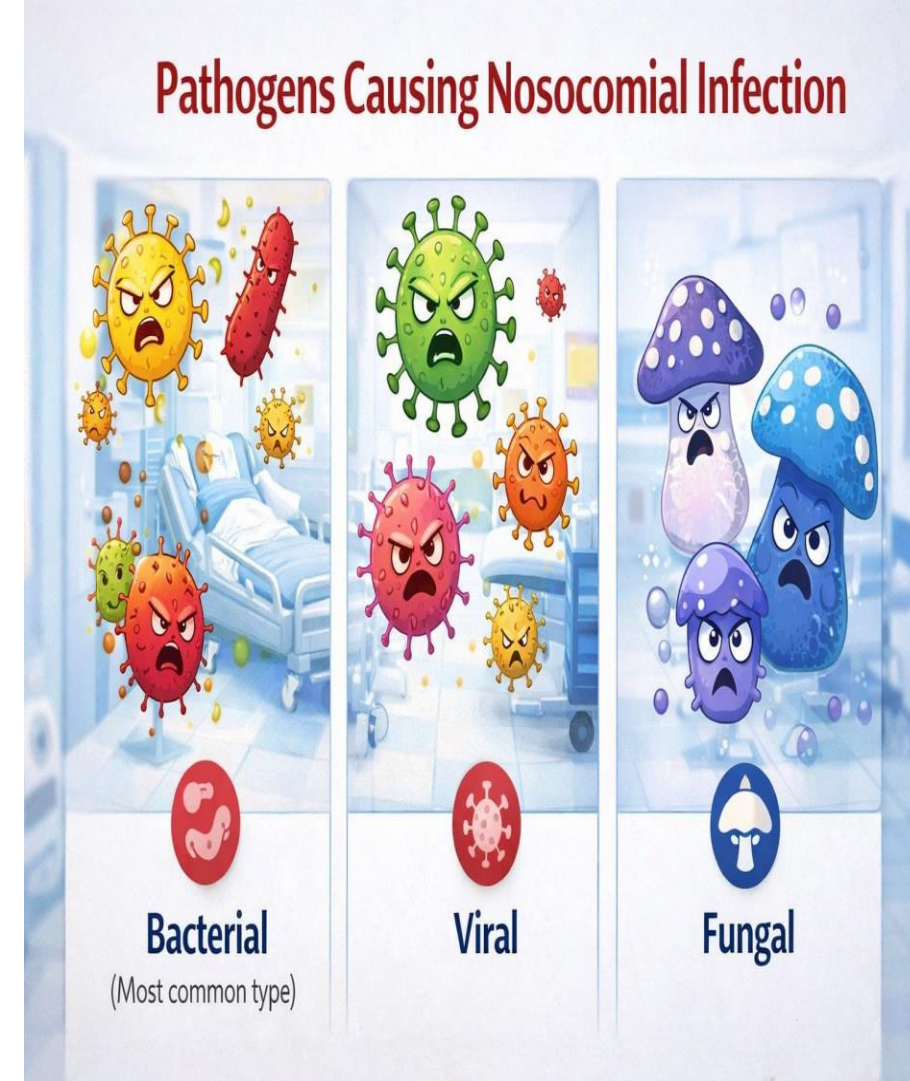
Types of pathogenic agents causing nosocomial infections.

The nature of pathogens causing nosocomial infections can be:

1. Bacterial (Most common type)

2. Viral

3. Fungal



Sources of Nosocomial Infection

Nosocomial infections may originate from:

1. **Healthcare personnel.**
2. **Contaminated instruments, bed linens, or airborne droplets.**
3. **The external environment.**
4. **Infected patients within the hospital, healthcare settings, or visitors.**
5. **The patient's own endogenous flora.**

 Click to watch this video and understand The Chain OF Infection



Susceptibility of Hospitalized Patients to Nosocomial Infection

Hospitalized patients are more prone to acquiring nosocomial infections due to the following factors:

- 1. Immunosuppressed condition:** Many patients undergo invasive medical procedures that breach the body's natural protective barriers such as the skin and mucous membranes.
- 2. Prolonged stay in intensive care units (ICUs):** Close contact with other critically ill patients who have weakened immune systems increases the risk of infection.
- 3. Extended use of antibiotics:** Long-term antibiotic therapy disrupts the normal microbial flora, allowing opportunistic pathogens to proliferate.

Penetration of Physical Barriers by Invasive Medical Procedures

The natural protective barriers of the body can be breached by several invasive medical procedures, increasing the risk of nosocomial infection:

1. **Surgical procedures:** Surgery disrupts the integrity of the skin, facilitating the entry of infectious agents.
2. **Mechanical ventilation:** Intubation and insertion of the endotracheal tube (ETT) damage the mucous membrane lining, allowing microorganisms to enter underlying tissues.
3. **Vascular catheterization:** Provides a direct pathway for pathogens to access the bloodstream.
4. **Urinary catheterization:** Damages the urethral mucous membrane, enabling pathogens to invade the tissues beneath it.

Sources of Bacterial Pathogens in Nosocomial Infections

Bacteria are the most common pathogens responsible for nosocomial infections and may originate from several sources:

- 1. Patient's normal (natural) flora:** These bacteria can cause infection when the patient's immune defenses are weakened.
- 2. Hospital environment:** Such as contaminated water, soil, and air.
- 3. Healthcare providers:** Who may transmit bacteria during patient care.

Common Bacterial Pathogens Causing Nosocomial Infections

The most bacterial agents responsible for nosocomial infections include:

1. Enterobacteriaceae: including *Klebsiella pneumoniae* and *E.coli*.
2. Gram-positive cocci: notably methicillin-resistant *Staphylococcus aureus* (**MRSA**) and vancomycin-resistant enterococci (**VRE**).

Viral Pathogens in Nosocomial Infections: Sources and Common Agents

Viruses represent an important cause of nosocomial infections and can be transmitted within healthcare settings through several routes:

Sources of transmission:

- **Unsafe injection practices.**
- **Surgical procedures from healthcare workers to patients.**
- **Use of contaminated blood or blood products.**
- **Respiratory secretions from healthcare providers and visitors.**

Common viral agents associated with nosocomial infections:

1. **Hepatitis B and C viruses**, commonly transmitted through unsafe injections and blood exposure.
2. Human Immunodeficiency Virus (**HIV**), transmitted via contaminated blood products.
3. **Influenza virus**, Rotavirus, Cytomegalovirus (CMV), and Herpes simplex virus.

Fungal Pathogens in Nosocomial Infections: Sources and Common Agents

Fungal infections acquired in healthcare settings may arise from the following sources:

1. **Environmental contamination**: Inhalation of **airborne fungal spores**, particularly during construction or renovation activities within healthcare facilities (e.g., **Aspergillus spp.**).
2. **Endogenous microflora**: Overgrowth of the patient's own fungi, especially **Candida albicans**, when immune defenses are compromised.

Common fungal pathogens associated with nosocomial infections:

Aspergillus spp.

Candida albicans

Types of Nosocomial Infections

Nosocomial infections may present in several clinical forms, classified into major and minor types.

Five major types:

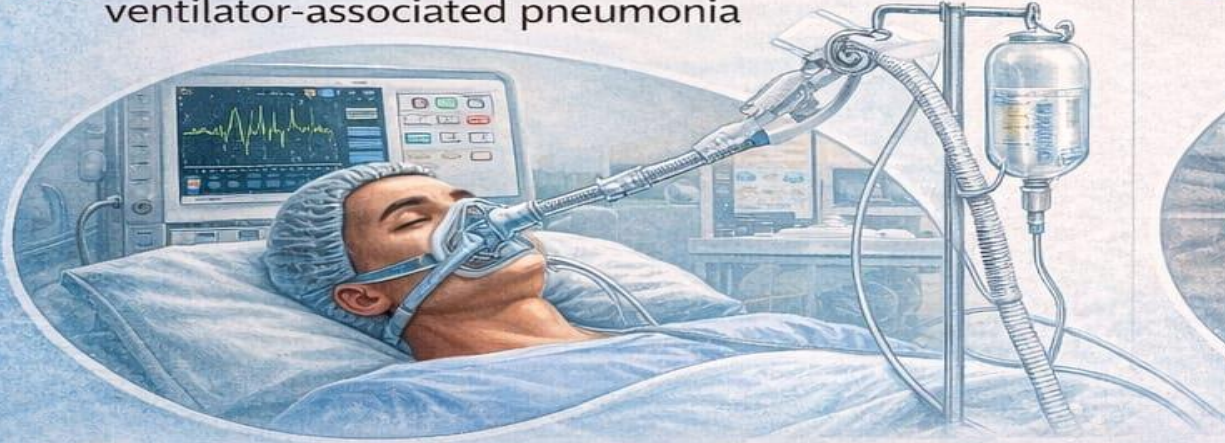
- 1. Hospital-acquired pneumonia (HAP) and ventilator-associated pneumonia (VAP)**
- 2. Urinary tract infection (UTI)**
- 3. Bloodstream infection (BSI)**
- 4. Surgical site infection (SSI)**

Types of Nosocomial Infections

Nosocomial infections may present in several clinical forms, classified into major and minor types.

Five Major Types

- 1 Hospital-acquired pneumonia (HAP) and ventilator-associated pneumonia**



- 2 Urinary tract infection (UTI)**



- 3 Bloodstream infection (BSI)**



- 4 Bloodstream infection (BSI)**

- 5 Surgical site infection (SSI)**



Other Minor Types

- 1 Gastroenteritis**



- 2 Puerperal fever**



Factors Increasing Infection Risk

1) Susceptible Host Factors (1

Extremes of **age** (newborns & elderly)

Impaired immunity:

Diabetes ,Cancer, HIV

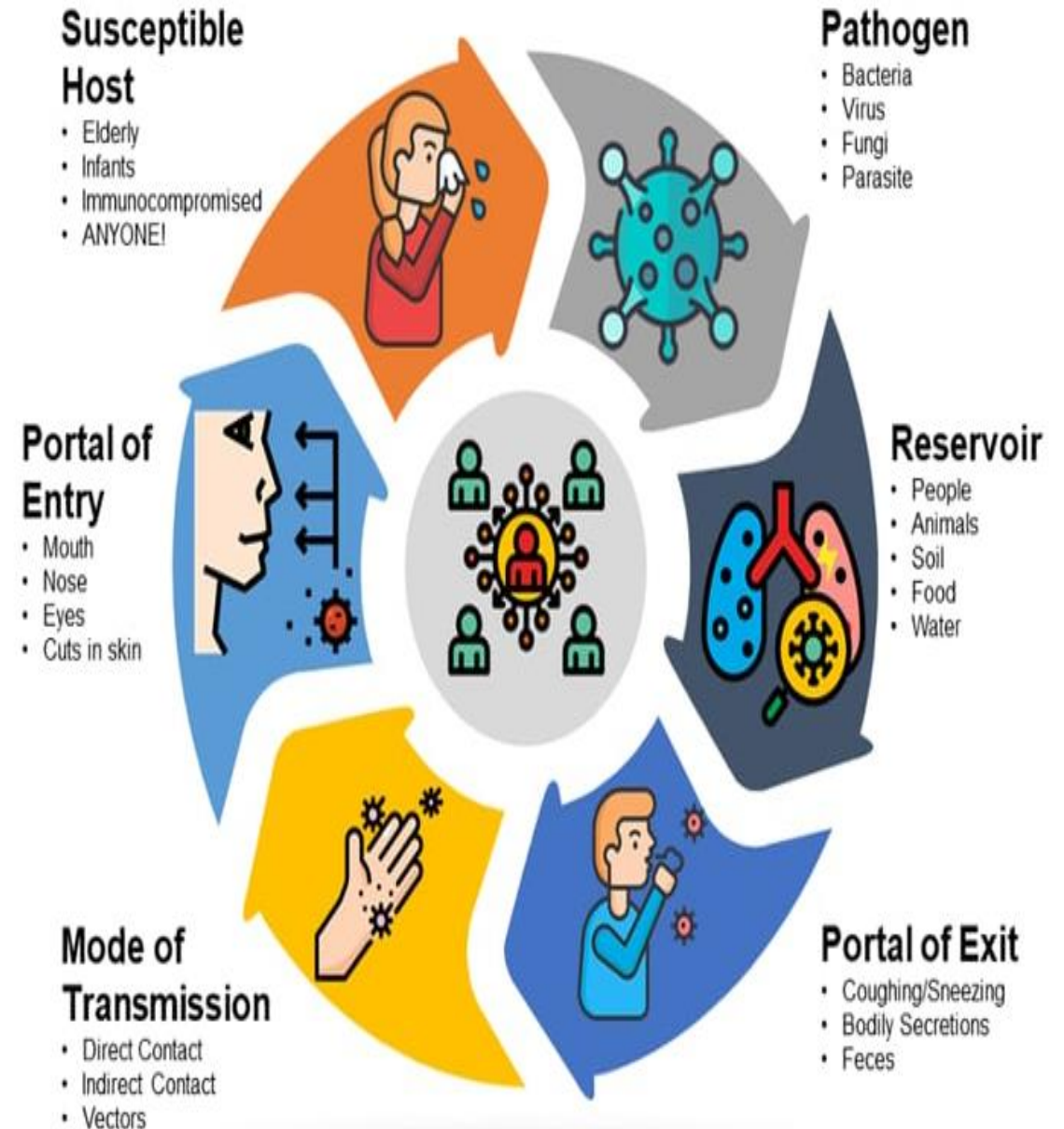
immunosuppressive therapy

Malnutrition

Chronic illnesses (renal, hepatic, cardiac)

Use of invasive devices:

Catheters, endoscopes, ventilators



2) Characteristics of Hospital Microorganisms

Highly pathogenic organisms

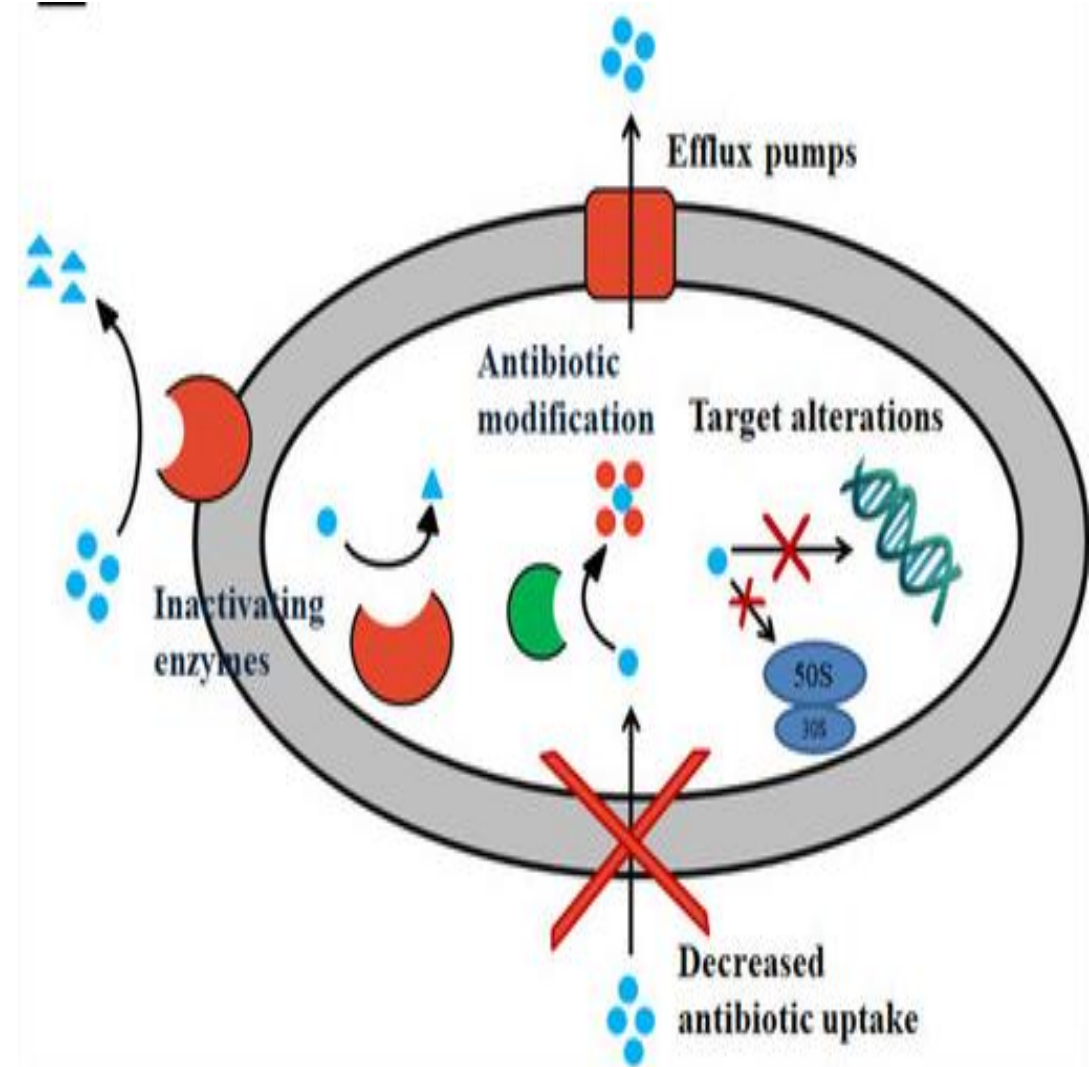
"Bacterial resistance arises through **genetic mutations** or the acquisition of **resistance genes**, allowing microorganisms to evade the effects of antibiotics."

Multidrug-resistant strains :

MRSA : Methicillin-resistant
Staphylococcus aureus

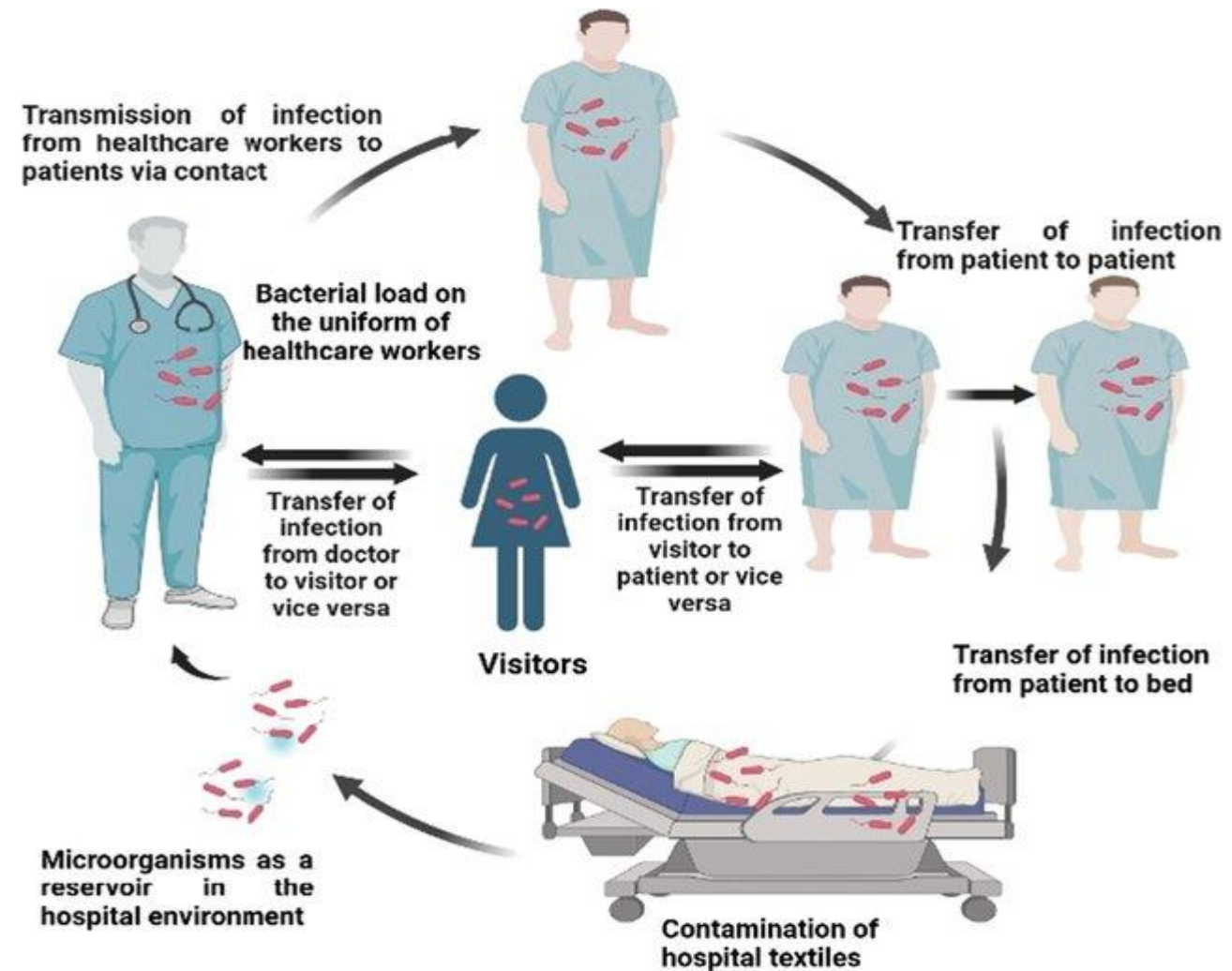
MRSE : Methicillin-resistant Staphylococcus
epidermidis

VRE : Vancomycin-resistant Enterococci



Hospital flora survive on:

- Surfaces
- Equipment
- Linen
- Medical devices
- These organisms spread easily and cause **severe infections**.



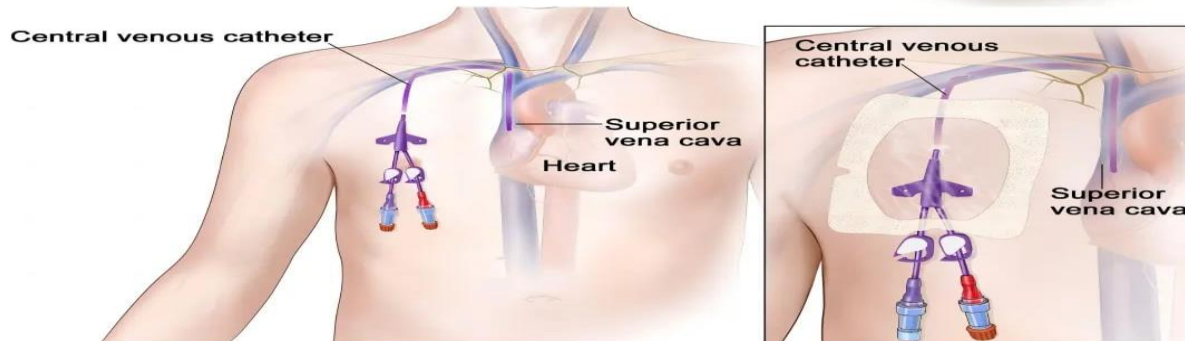
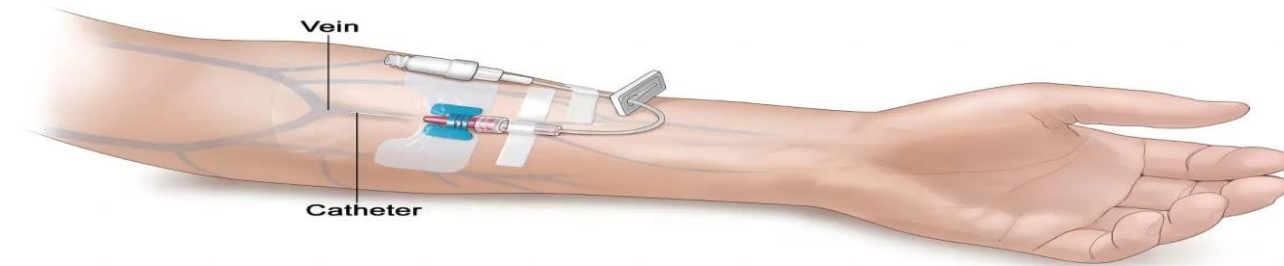
Created with BioRender.com

- Hospital microorganisms survive on multiple surfaces and are easily transferred between patients, healthcare workers, and visitors, leading to continuous cycles of contamination and infection.

Invasive procedures increase the risk:

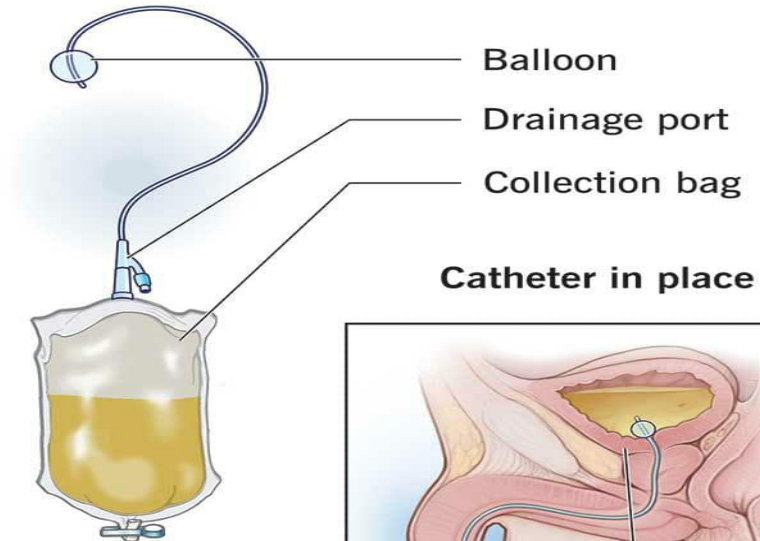
1. Urinary catheters

2. IV lines

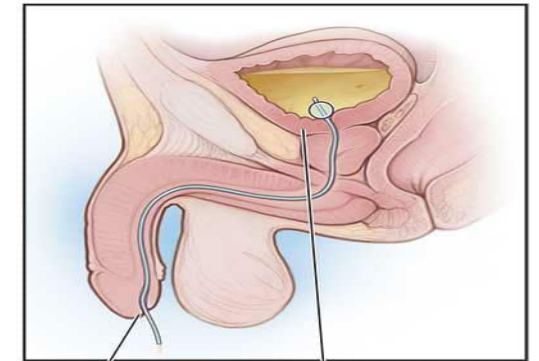


Foley catheter

Foley catheter



Catheter in place

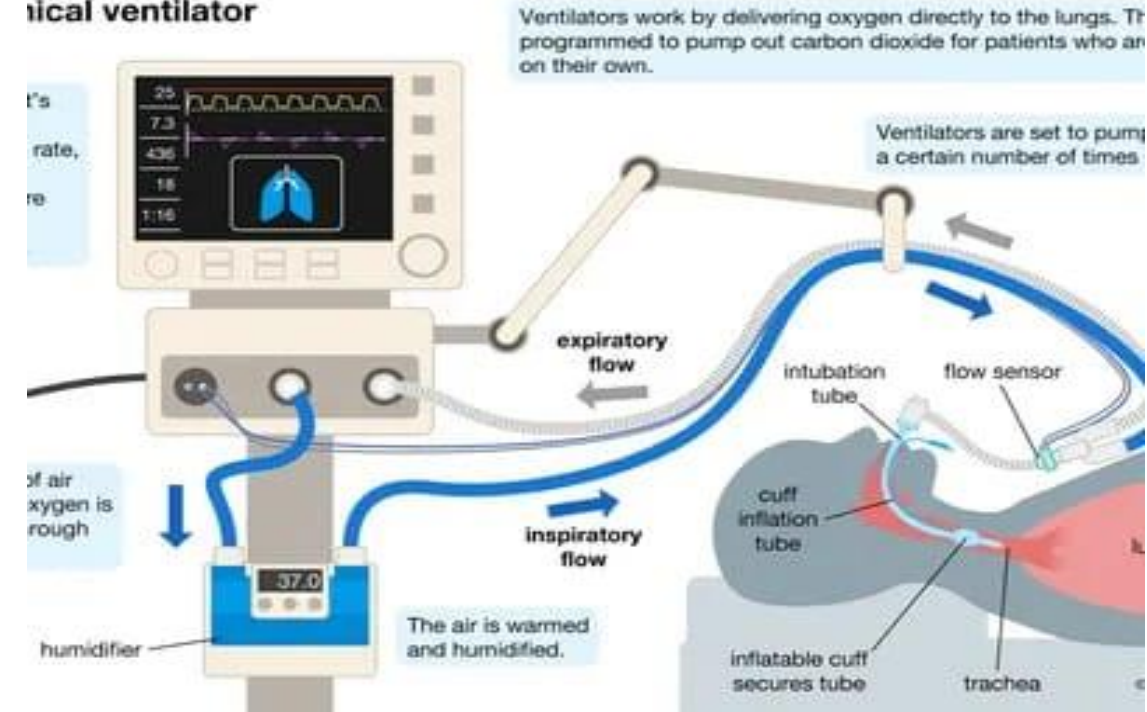


Cleveland
Clinic
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3.Mechanical ventilation

4.Surgical procedures

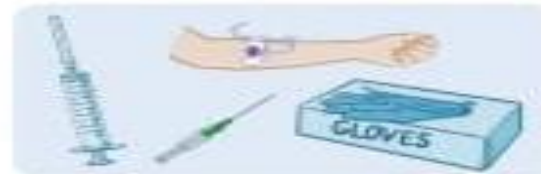
ical ventilator



SURGICAL ASEPSIS - AVOID EXPOSING CLIENT to MICROORGANISMS



USE of DISPOSABLE STERILE SUPPLIES



USE of REUSABLE STERILE EQUIPMENT



STERILE TECHNIQUE - ↓ CHANCE of EXPOSING CLIENT to BACTERIA



SPECIFIC RULES when USING & APPLYING STERILE EQUIPMENT



Prevention and Control of Nosocomial Infections

- 1. Hand hygiene:** Wash or disinfect hands before and after touching each patient.
- 2. Non-critical equipment:** Do not share equipment like stethoscopes between patients unless cleaned first.
- 3. Periurethral cleansing:** Clean the area around the urinary opening before inserting a urinary catheter.
- 4. Urinary catheters:** Use urinary catheters only when needed and remove them as soon as possible.

Prevention and Control of Nosocomial Infections

6. **Antibiotic therapy:** Use the right antibiotic for the specific germ causing the infection.
7. **Colonized or infected ICU patients:** Keep infected or colonized ICU patients in isolation rooms to stop infection spread.
8. **Vaginal examinations:** Avoid unnecessary vaginal examinations and douching in late pregnancy.
9. **Existing infections:** Treat any infection before delivery, especially vaginal infections.
10. **Management of anemia and diabetes:** Control anemia and diabetes well to reduce the risk of infection.

MCQs

Question

1

Which factor is most related to opportunistic pathogen growth?

- A** Normal flora disruption
- B** Short hospital stay
- C** Strong immune defense
- D** No antibiotic exposure

Question

2

Before inserting a urinary catheter, the healthcare provider should:

- A** Skip cleaning if the patient looks healthy
- B** Clean the periurethral area with antiseptic
- C** Use the catheter without gloves
- D** Keep the catheter as long as possible

MCQs

3. Nosocomial infections are also known as:

- ☐ A Genetic infections
- ☐ B Hospital-acquired infections
- ☐ C Nutritional infections
- ☐ D Autoimmune infections

4. Nosocomial infections usually develop after:

- ☐ A 6 hours of admission
- ☐ B 12 hours of admission
- ☐ C 24 hours of admission
- ☐ D 48 hours or more after admission

THANK YOU

Infection Control Saves Lives

Dr. Rania Hamed

